

Adam Doupe

P.O. Box 878809
Tempe, AZ 85287-8809
📞 480-727-5471
✉️ doupe@asu.edu
🌐 adamdoupe.com

Education

- 2010–2014 **PhD in Computer Science**, *University of California, Santa Barbara*
Advisors: Giovanni Vigna and Christopher Kruegel
Thesis: Advanced Automated Web Application Vulnerability Analysis
- 2008–2009 **Master's Degree in Computer Science**, *University of California, Santa Barbara*
Advisor: Giovanni Vigna
Topic: Black Box Web Vulnerability Scanners
- 2004–2008 **Bachelor's Degree in Computer Science with Honors**, *University of California, Santa Barbara*

Professional Appointments

- 2020–Present **Associate Professor**, *Arizona State University*
- 2021–Present **Director**, *Center for Cybersecurity and Trusted Foundations*, Arizona State University
- 2020–2021 **Acting Director**, *Center for Cybersecurity and Trusted Foundations*, Arizona State University
- 2017–2020 **Associate Director**, *Center for Cybersecurity and Trusted Foundations*, Arizona State University
- 2014–2020 **Assistant Professor**, *Arizona State University*

Research Funding

27. DARPA 2024–2026. *American Cybersecurity Education (ACE) Institute*, \$4,462,558, co-PI (w/ Yan Shoshitaishvili as PI and Fish Wang and Tiffany Bao as co-PIs), share \$1,115,639.
26. NSA 2024–2029. *STEEDDEMON: Human-assisted Cyber Reasoning Systems*, \$2,496,407, co-PI (w/ Yan Shoshitaishvili as PI and Chitta Baral as co-PI), share \$998,562.
25. Stony Brook University 2024–2027. *ONR: Tools and Techniques for Characterizing Integrity in DevOps Technologies*, \$300,001, PI, share \$300,001.
24. DARPA 2024. *STTR: SENSEI: Scaling Education Nationwide for Security Expertise and Impact*, \$82,234, co-PI (w/ Yan Shoshitaishvili as PI).
23. ARPA-H 2024–2024. *RxCRS: Reliable and eXplainable Cyber Reasoning System for Digital Health Security*, \$10,186,189, co-PI (w/ Fish Wang as PI and Tiffany Bao, Kenneth Buetow, Stephanie Forrest, Duong Nguyen, Yan Shoshitaishvili, and Ni Trieu as co-PIs), share \$1,833,514.

22. DARPA 2023–2027. *SENPAL: Strategic Exploration, Navigation, and Patching of Abstracted Integration*, \$7,960,690, co-PI (w/ Tiffany Bao as PI and Sandeep Gupta, Zilin Jiang, Giulia Pedrielli, Yan Shoshitaishvili, and Ruoyu Wang as co-PIs), share \$1,990,172.
21. NSA 2023–2026. *SoS: Leveraging Machine Learning for Binary Software Understanding*, \$748,949, co-PI (w/ Yan Shoshitaishvili as PI and Chitta Baral as co-PI), share \$441,678.
20. ARO 2023–2024. *DURIP: Next Generation Machine Learning Platform for Advanced Vulnerability Analysis and Reverse Engineering*, \$209,283, co-PI (w/ Yan Shoshitaishvili as PI and Tiffany Bao as co-PI), share \$52,320.
19. Ripple 2022–2023. *Ripple Impact Fund*, \$34,999, PI.
18. PayPal 2022–2023. *Protecting the online ecosystem through proactive cybercrime reporting*, \$100,001, PI.
17. NSA 2022–2024. *Human-Assisted Cyber Reasoning Systems Cont.*, \$2,178,001, co-PI (w/ Yan Shoshitaishvili PI and Chitta Baral as co-PI), share \$990,000.
16. DARPA 2022–2026. *EURYALE: Combating Emergent Execution with UTE*, \$3,768,945, PI, (w/ Fish Wang, Yan Shoshitaishvili, Tiffany Bao, Stephanie Forrest, and Giulia Pedrielli as co-PIs), share \$1,130,683.
15. ONR 2020–2023. *Tools and Techniques to Improve the Granularity and Usability of Web Application*, \$1,599,983, PI.
14. DARPA 2020–2024. *DARPA AMP: VOLT: A Viscous, Orchestrated Lifting and Translation Framework*, \$6,678,767, co-PI (w/ Fish Wang PI and Tiffany Bao, Georgios Fainekos, Stephanie Forrest, Sandeep Gupta, and Yan Shoshitaishvili as co-PIs), share \$1,001,815.
13. NSA 2019–2021. *Human-Assisted Cyber Reasoning Systems*, \$1,299,053, co-PI (w/ Yan Shoshitaishvili PI and Tiffany Bao as co-PI), share \$436,703.
12. IBM 2019–2022. *DARPA CHASE: Hardened Orchestrated Response for Uncertain Settings (HORUS)*, \$674,083, PI (w/ Subbarao Kambhampati co-PI), share \$337,041.
11. DARPA 2019–2022. *DARPA CHERS: Cognitive Human Enhancements for Cyber Reasoning Systems*, \$11,730,557, co-PI (w/ Fish Wang PI and Yan Shoshitaishvili PI and Tiffany Bao, Chitta Baral, and Stephanie Forrest as co-PIs), share \$1,524,972.
10. DARPA 2018–2022. *DARPA HACCS: Puppeteer—Achieving Remote Control Over Complex Systems*, \$7,474,245, co-PI (w/ Yan Shoshitaishvili PI), share \$1,868,561.
9. NSF 2018–2023. *NSF ADVANCE: ASU ADVANCE Institutional Transformation*, \$2,999,743, Co-Investigator (w/ Elizabeth Wentz as PI), share \$29,997.
8. NSF 2017–2022. *NSF-SFS: Arizona Cyber Defense Scholarship*, \$3,997,784, co-PI (w/ Gail-Joon Ahn PI and Steven Yau, Ziming Zhao, and Dijiang Huang co-PIs), share \$399,778.

7. NSF 2017–2021. *SaTC: CORE: Medium: Collaborative: Taming Web Content Through Automated Reduction in Browser Functionality*, \$406,080, PI.
6. Soongsil University 2017–2024. *Automatic Deep Malware Analysis Technology for Cyber Threat Intelligence*, \$708,160, co-PI (w/ Gail-Joon Ahn PI), share \$283,264.
5. ONR 2017–2019. *XS-Shredder: A Cross-Layer Framework for Removing Code Bloat in Web Applications*, \$1,230,547, PI.
4. NSF 2017–2022. *CAREER: Next Generation Black-Box Web Application Vulnerability Analysis*, \$416,585, PI.
3. NSF 2016–2019. *Collaborative Research: CICI: Secure and Resilient Architecture: SciGuard: Building a Security Architecture for Science DMZ based on SDN and NFV Technologies*, \$499,484, co-PI (w/ Gail-Joon Ahn PI and Dijiang Huang co-PI), share \$149,845.
2. NSF 2016–2018. *EDU: Collaborative: Educating the Security Workforce through On-Demand Live Competitions*, \$154,643, PI (w/ Gail-Joon Ahn co-PI), share \$77,321.
1. ONR 2015. *Academic Survey of the Cyber Resiliency of the Total Embedded Systems Platform*, \$49,998, PI.

Honors and Awards

- 2021 **Best Student Paper Award**, *IEEE Symposium on Security and Privacy*
- 2020 **Second Place Internet Defense Prize**, *Facebook*
- 2020 **Distinguished Paper Award**, *USENIX Security Symposium (USENIX)*
- 2019 **Distinguished Paper Award**, *USENIX Security Symposium (USENIX)*
- 2019 **Fulton Schools of Engineering Best Teacher Award—Top 5%**, *Arizona State University*
- 2018 **Fulton Schools of Engineering Best Teacher Award**, *Arizona State University*
- 2017 **Fulton Outstanding Assistant Professor**, *Arizona State University*
- 2017 **Fulton Schools of Engineering Best Teacher Award**, *Arizona State University*
- 2017 **NSF CAREER Award**
- 2016 **Best Paper Award**, *Toward Authenticated Caller ID Transmission: The Need for a Standardized Authentication Scheme in Q.731.3 Calling Line Identification Presentation*, ITU Kaleidoscope
- 2016 **Fulton Schools of Engineering Best Teacher Award—Top 5%**, *Arizona State University*
- 2015 **Best Paper Award**, *A Large-Scale Study of Mobile Web App Security*, *Mobile Security Technologies Workshop*
- 2013 **Outstanding Publication Award in Computer Science**, *Enemy of the State: A State-Aware Black-Box Web Vulnerability Scanner*, *University of California, Santa Barbara*

- 2011 **Outstanding Teaching Assistant Award**, CS 279—(Graduate) *Network Security and Intrusion Detection*, University of California, Santa Barbara
- 2010 **Gold Star Award**, *Microsoft*
- 2009 **Outstanding Teaching Assistant Award**, CS 177—*Introduction to Computer Security*, University of California, Santa Barbara

Publications

Students advised, co-advised, or mentored are underlined, undergraduate students contain an asterisk next to their name, and conference acceptance rates are provided when available. First authorship indicates primary contribution, unless otherwise noted.

Journal Papers (refereed)

4. Pemma Reiter, Hui Jun Tay, Westley Weimer, **Adam Doupé**, Ruoyu Wang, and Stephanie Forrest, "Automatically Mitigating Vulnerabilities in Binary Programs via Partially Recompileable Decompilation," in *IEEE Transactions on Dependable and Secure Computing*, October, 2024.
3. Edward Schrom, Ann Kinzig, Stephanie Forrest, Andrea L. Graham, Simon A. Levin, Carl T. Bergstrom, Carlos Castillo-Chavez, James P. Collins, Rob J. de Boer, **Adam Doupé**, Roya Ensafi, Stuart Feldman, Bryan T. Grenfell, J. Alex Halderman, Silvie Huijben, Carlo Maley, Melanie Moses, Alan S. Perelson, Charles Perrings, Joshua Plotkin, Jennifer Rexford, and Mohit Tiwari, "Challenges in cybersecurity: Lessons from biological defense systems," in *Mathematical Biosciences*, June, 2023.
2. Huahong Tu, **Adam Doupé**, Ziming Zhao, and Gail-Joon Ahn, "Toward Standardization of Authenticated Caller ID Transmission," in *IEEE Communications Standards Magazine*, 1(3), pp. 30-36, October 2017.
1. Sai Prashanth Chandramouli, Ziming Zhao, **Adam Doupé**, and Gail-Joon Ahn, "E-mail Header Injection Vulnerabilities," in *it - Information Technology*, 59(2), pp. 67-72, March 2017.

Conference Papers with Proceedings (refereed)

76. Marzieh Bitaab, Alireza Karimi, Zhuoer Lyu, Ahmadreza Mosallanezhad, Adam Oest, Ruoyu Wang, Tiffany Bao, Yan Shoshitaishvili, and **Adam Doupé**, "ScamNet: Toward Explainable Large Language Model-based Fraudulent Shopping Website Detection," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, February, 2025.
75. Marzieh Bitaab, Alireza Karimi, Zhuoer Lyu, Adam Oest, Dhruv Kuchhal, Muhammad Saad, Gail-Joon Ahn, Ruoyu Wang, Tiffany Bao, Yan Shoshitaishvili, and **Adam Doupé**, "ScamMagnifier: Piercing the Veil of Fraudulent Shopping Website Campaigns," in *Proceedings of the Symposium on Network and Distributed System Security (NDSS)*, February, 2025.
74. Faezeh Kalantari, Mehrnoosh Zaeifi, Yeganeh Safaei, Marzieh Bitaab, Adam Oest, Gianluca Stringhini, Yan Shoshitaishvili, and **Adam Doupé**, "Browser Polygraph: Efficient Deployment of Coarse-Grained Browser Fingerprints for Web-Scale Detection of Fraud Browsers," in *Proceedings of the ACM Internet Measurement Conference (IMC)*, November, 2024.

73. Arvind S Raj, Wil Gibbs, Fangzhou Dong, Jayakrishna Vadayath, Michael Tompkins, Steven Wirsz, Yibo Liu, Zhenghao Hu, Chang Zhu, Gokulkrishna Praveen Menon, Brendan Dolan-Gavitt, **Adam Doupé**, Ruoyu Wang, Yan Shoshitaishvili, and Tiffany Bao, “Fuzz to the Future: Uncovering Occluded Future Vulnerabilities via Robust Fuzzing,” in *Proceedings of the ACM Conference on Computer and Communications Security (CCS)*, October, 2024.
72. Zhibo Sun, Faris Bugra Kokulu, Penghui Zhang, Adam Oest, Gianluca Stringhini, Tiffany Bao, Ruoyu Wang, Yan Shoshitaishvili, **Adam Doupé**, and Gail-Joon Ahn, “From Victims to Defenders: An Exploration of the Phishing Attack Reporting Ecosystem,” in *Proceedings of the International Symposium on Research in Attacks, Intrusions and Defenses (RAID)*, September, 2024.
71. Zion Leonahenahe Basque, Ati Priya Bajaj, Wil Gibbs, Jude O’Kain, Derron Miao, Tiffany Bao, **Adam Doupé**, Yan Shoshitaishvili, and Ruoyu Wang, “Ahoy SAILR! There is No Need to DREAM of C: A Compiler-Aware Structuring Algorithm for Binary Decompilation,” in *Proceedings of the USENIX Security Symposium (USENIX)*, August, 2024.
70. Ananta Soneji, Vaughn Hamilton, **Adam Doupé**, Allison McDonald, and Elissa M. Redmiles, ““I feel physically safe but not politically safe”: Understanding the Digital Threats and Safety Practices of OnlyFans Creators,” in *Proceedings of the USENIX Security Symposium (USENIX)*, August, 2024.
69. Wil Gibbs, Arvind S Raj, Jayakrishna Vadayath, Hui Jun Tay, Justin Miller, Akshay Ajayan, Zion Leonahenahe Basque, Audrey Dutcher, Fangzhou Dong, Xavier J. Maso, Giovanni Vigna, Christopher Kruegel, **Adam Doupé**, Yan Shoshitaishvili, and Ruoyu Wang, “Operation Mango: Scalable Discovery of Taint-Style Vulnerabilities in Binary Firmware Services,” in *Proceedings of the USENIX Security Symposium (USENIX)*, August, 2024.
68. Eric Olsson, Benjamin Eriksson, **Adam Doupé**, and Andrei Sabelfeld, “Spider-Scents: Grey-box Database-aware Web Scanning for Stored XSS,” in *Proceedings of the USENIX Security Symposium (USENIX)*, August, 2024.
67. Ziyi Guo, Dang K Le, Zhenpeng Lin, Kyle Zeng, Ruoyu Wang, Tiffany Bao, Yan Shoshitaishvili, **Adam Doupé**, and Xinyu Xing, “Take a Step Further: Understanding Page Spray in Linux Kernel Exploitation,” in *Proceedings of the USENIX Security Symposium (USENIX)*, August, 2024.
66. Chang Zhu, Ziyang Li, Anton Xue, Ati Priya Bajaj, Wil Gibbs, Yibo Liu, Rajeev Alur, Tiffany Bao, Hanjun Dai, **Adam Doupé**, Mayur Naik, Yan Shoshitaishvili, Ruoyu Wang, and Aravind Machiry, “TYGR: Type Inference on Stripped Binaries using Graph Neural Networks,” in *Proceedings of the USENIX Security Symposium (USENIX)*, August, 2024.
65. Vivin Paliath, Erik Trickel, Tiffany Bao, Ruoyu Wang, **Adam Doupé**, and Yan Shoshitaishvili, “SandPuppy: Deep-State Fuzzing Guided by Automatic Detection of State-Representative Variables,” in *Proceedings of the Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA)*, July, 2024.
64. Rana Pourmohamad, Steven Wirsz, Adam Oest, Tiffany Bao, Yan Shoshitaishvili, Ruoyu Wang, **Adam Doupé**, and Rida A. Bazzi, “Deep Dive into Client-Side Anti-Phishing: A Longitudinal Study

Bridging Academia and Industry," in *Proceedings of the ACM Symposium on Information, Computer and Communications Security (AsiaCCS)*, July, 2024.

63. Mehrnoosh Zaeifi, Faezeh Kalantari, Adam Oest, Zhibo Sun, Gail-Joon Ahn, Yan Shoshitaishvili, Tiffany Bao, Ruoyu Wang, and **Adam Doupé**, "Nothing Personal: Understanding the Spread and Use of Personally Identifiable Information in the Financial Ecosystem," in *Proceedings of the ACM on Conference on Data and Application Security and Privacy (CODASPY)*, June, 2024.
62. Kuntal Kumar Pal*, Ati Priya Bajaj*, Pratyay Banerjee, Audrey Dutcher, Mutsumi Nakamura, Zion Leonahenahe Basque, Himanshu Gupta, Saurabh Arjun Sawant, Ujjwala Anantheswaran, Yan Shoshitaishvili, **Adam Doupé**, Chitta Baral, and Ruoyu Wang, ""Len or index or count, anything but v1": Predicting Variable Names in Decompilation Output with Transfer Learning," in *Proceedings of the IEEE Symposium on Security and Privacy*, May, 2024.
*Co-first authors
61. Irina Ford, Ananta Soneji, Faris Bugra Kokulu, Jayakrishna Vadayath, Zion Leonahenahe Basque, Gaurav Vipat, **Adam Doupé**, Ruoyu Wang, Gail-Joon Ahn, Tiffany Bao, and Yan Shoshitaishvili, ""Watching over the shoulder of a professional": Why Hackers Make Mistakes and How They Fix Them," in *Proceedings of the IEEE Symposium on Security and Privacy*, May, 2024.
60. Kyle Zeng, Zhenpeng Lin, Kangjie Lu, Xinyu Xing, Ruoyu Wang, **Adam Doupé**, Yan Shoshitaishvili, and Tiffany Bao, "RetSpill: Igniting User-Controlled Data to Burn Away Linux Kernel Protections," in *Proceedings of the ACM Conference on Computer and Communications Security (CCS)*, November 2023.
59. Hui Jun Tay, Kyle Zeng, Jayakrishna Vadayath, Arvind S Raj, Audrey Dutcher, Tejesh Reddy, Wil Gibbs, Zion Leonahenahe Basque, Fangzhou Dong, Zack Smith, **Adam Doupé**, Tiffany Bao, Yan Shoshitaishvili, and Ruoyu Wang, "Greenhouse: Single-Service Rehosting of Linux-Based Firmware Binaries in User-Space Emulation," in *Proceedings of the USENIX Security Symposium (USENIX)*, August, 2023.
58. Jaejong Baek, Pradeep Kumar Duraisamy Soundrapandian, Sukwha Kyung, Ruoyu Wang, Yan Shoshitaishvili, **Adam Doupé**, and Gail-Joon Ahn, "Targeted Privacy Attacks by Fingerprinting Mobile Apps in LTE Radio Layer," in *Proceedings of the IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, June, 2023.
57. Marzieh Bitaab, Haehyun Cho, Adam Oest, Zhuoer Lyu, Wei Wang, Jorij Abraham, Ruoyu Wang, Tiffany Bao, Yan Shoshitaishvili, and **Adam Doupé**, "Beyond Phish: Toward Detecting Fraudulent e-Commerce Websites at Scale," in *Proceedings of the IEEE Symposium on Security and Privacy*, May 2023.
56. Erik Trickel, Fabio Pagani, Chang Zhu, Lukas Dresel, Giovanni Vigna, Christopher Kruegel, Ruoyu Wang, Tiffany Bao, Yan Shoshitaishvili, and **Adam Doupé**, "Toss a Fault to Your Witcher: Applying Grey-box Coverage-Guided Mutational Fuzzing to Detect SQL and Command Injection Vulnerabilities," in *Proceedings of the IEEE Symposium on Security and Privacy*, May, 2023.
55. Penghui Zhang, Zhibo Sun, Sukwha Kyung, Hans Walter Behrens, Zion Leonahenahe Basque, Haehyun Cho, Adam Oest, Ruoyu Wang, Tiffany Bao, Yan Shoshitaishvili, Gail-Joon Ahn, and **Adam**

- Doupé**, “I’m Spartacus, No, I’m Spartacus: Proactively Protecting Users from Phishing by Intentionally Triggering Cloaking Behavior,” in *Proceedings of the ACM Conference on Computer and Communications Security (CCS)*, November, 2022.
54. Faezeh Kalantari, Mehrnoosh Zaeifi, Tiffany Bao, Ruoyu Wang, Yan Shoshitaishvili, and **Adam Doupé**, “Context-Auditor: Context-sensitive Content Injection Mitigation,” in *Proceedings of the International Symposium on Research in Attacks, Intrusions and Defenses (RAID)*, October, 2022.
 53. Siddhant Bhambri*, Purv Chauhan*, Frederico Araujo, **Adam Doupé**, and Subbarao Kambhampati, “Using Deception in Markov Game to Understand Adversarial Behaviors through a Capture-The-Flag Environment,” in *Proceedings of the Conference on Decision and Game Theory for Security (GameSec)*, October, 2022.
*Co-first authors
 52. Soroush Karami*, Faezeh Kalantari*, Mehrnoosh Zaeifi, Xavier J. Maso, Erik Trickel, Panagiotis Ilia, Yan Shoshitaishvili, **Adam Doupé**, and Jason Polakis, “Unleash the Simulacrum: Shifting Browser Realities for Robust Extension-Fingerprinting Prevention,” in *Proceedings of the USENIX Security Symposium (USENIX)*, August, 2022.
*Co-first authors
 51. Jayakrishna Vadayath, Moritz Eckert, Kyle Zeng, Nicolaas Weideman, Gokulkrishna Praveen Menon, Yanick Fratantonio, Davide Balzarotti, **Adam Doupé**, Tiffany Bao, Ruoyu Wang, Christophe Hauser, and Yan Shoshitaishvili, “Arbiter: Bridging the Static and Dynamic Divide in Vulnerability Discovery on Binary Programs,” in *Proceedings of the USENIX Security Symposium (USENIX)*, August, 2022.
 50. Kyle Zeng*, Yueqi Chen*, Haehyun Cho, Xinyu Xing, **Adam Doupé**, Yan Shoshitaishvili, and Tiffany Bao, “Playing for K(H)eaps: Understanding and Improving Linux Kernel Exploit Reliability,” in *Proceedings of the USENIX Security Symposium (USENIX)*, August, 2022.
*Co-first authors
 49. Pemma Reiter*, Antonio M. Espinoza*, **Adam Doupé**, Ruoyu Wang, Westley Weimer, and Stephanie Forrest, “Improving Source-Code Representations to Enhance Search-based Software Repair,” in *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO)*, July, 2022.
*Co-first authors
 48. Ananta Soneji, Faris Bugra Kokulu, Carlos E. Rubio-Medrano, Tiffany Bao, Ruoyu Wang, Yan Shoshitaishvili, and **Adam Doupé**, “‘Flawed, but like democracy we don’t have a better system’: The Experts’ Insights on the Peer Review Process of Evaluating Security Papers,” in *Proceedings of the IEEE Symposium on Security and Privacy*, May, 2022.
 47. Rock Stevens*, Faris Bugra Kokulu*, **Adam Doupé**, and Michelle Mazurek, “Above and Beyond: Organizational Efforts to Complement U.S. Digital Security Compliance Mandates,” in *Proceedings of the Symposium on Network and Distributed System Security (NDSS)*, February, 2022.
*Co-first authors
 46. Haehyun Cho, Jinbum Park, Adam Oest, Tiffany Bao, Ruoyu Wang, Yan Shoshitaishvili, **Adam Doupé**, and Gail-Joon Ahn, “ViK: Practical Mitigation of Temporal Memory Safety Violations

through Object ID Inspection,” in *Proceedings of the ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, February, 2022.

- 45. Zhibo Sun, Adam Oest, Penghui Zhang, Carlos E. Rubio-Medrano, Tiffany Bao, Ruoyu (Fish) Wang, Ziming Zhao, Yan Shoshitaishvili, **Adam Doupé**, and Gail-Joon Ahn, “Having Your Cake and Eating It: An Analysis of Concession-Abuse-as-a-Service,” in *Proceedings of the USENIX Security Symposium (USENIX)*, August, 2021.
- 44. Penghui Zhang, Adam Oest, Haehyun Cho, Zhibo Sun, RC Johnson, Brad Wardman, Shaown Sarker, Alexandros Kapravelos, Tiffany Bao, Ruoyu (Fish) Wang, Yan Shoshitaishvili, **Adam Doupé**, and Gail-Joon Ahn, “CrawlPhish: Large-scale Analysis of Client-side Cloaking Techniques in Phishing,” in *Proceedings of the IEEE Symposium on Security and Privacy*, May, 2021.

Best Student Paper Award

- 43. Jaswant Pakki, Yan Shoshitaishvili, Ruoyu Wang, Tiffany Bao, and **Adam Doupé**. “Everything You Ever Wanted to Know About Bitcoin Mixers (But Were Afraid to Ask),” in *Proceedings of the Financial Cryptography and Data Security Conference*, March, 2021.
- 42. Sung Ta Dinh, Haehyun Cho, Kyle Martin, Adam Oest, Kyle Zeng, Alexandros Kapravelos, Gail-Joon Ahn, Tiffany Bao, Ruoyu Wang, **Adam Doupé**, and Yan Shoshitaishvili, “Favocado: Fuzzing the Binding Code of JavaScript Engines Using Semantically Correct Test Cases,” in *Proceedings of the ISOC Network and Distributed System Security Symposium (NDSS)*, February, 2021.
- 41. Frederico Araujo, Sailik Sengupta, Jiyong Jang, **Adam Doupé**, Kevin W. Hamlen, and Subbarao Kambhampati, “Software Deception Steering through Version Emulation,” in *Proceedings of the Hawaii International Conference on System Sciences*, Online, January, 2021.
- 40. Pradeep Kumar Duraisamy Soundrapandian, Tiffany Bao, Jaejong Baek, Yan Shoshitaishvili, **Adam Doupé**, Ruoyu Wang, and Gail-Joon Ahn, “MuTent: Dynamic Android Intent Protection with Ownership-Based Key Distribution and Security Contracts,” in *Proceedings of the Hawaii International Conference on System Sciences*, Online, January, 2021.
- 39. Marzieh Bitaab, Haehyun Cho, Adam Oest, Penghui Zhang, Zhibo Sun, Rana Pourmohamad, Dowoon Kim, Tiffany Bao, Ruoyu Wang, Yan Shoshitaishvili, **Adam Doupé**, and Gail-Joon Ahn, “Scam Pandemic: How Attackers Exploit Public Fear through Phishing,” in *Proceedings of the Symposium on Electronic Crime Research (eCrime)*, November, 2020.
- 38. Efrén López Morales, Carlos E. Rubio-Medrano, **Adam Doupé**, Yan Shoshitaishvili, Ruoyu Wang, Tiffany Bao, and Gail-Joon Ahn, “HoneyPLC: A Next-Generation Honeypot for Industrial Control Systems,” in *Proceedings of the ACM Conference on Computer and Communications Security (CCS)*, November, 2020. (16.9% acceptance rate.)
- 37. Adam Oest, Penghui Zhang, Brad Wardman, Eric Nunes, Jakub Burgis, Ali Zand, Kurt Thomas, **Adam Doupé**, and Gail-Joon Ahn, “Sunrise to Sunset: Analyzing the End-to-end Life Cycle and Effectiveness of Phishing Attacks at Scale,” in *Proceedings of the USENIX Security Symposium (USENIX)*, Boston, MA, August, 2020. (16.1% acceptance rate.)

Distinguished Paper Award
Second Place Internet Defense Prize

36. Adam Oest, Yeganeh Safaei, Penghui Zhang, Brad Wardman, Kevin Tyers, Yan Shoshitaishvili, **Adam Doupé**, and Gail-Joon Ahn, "PhishTime: Continuous Longitudinal Measurement of the Effectiveness of Anti-phishing Blacklists," in *Proceedings of the USENIX Security Symposium (USENIX)*, Boston, MA, August, 2020. (16.1% acceptance rate.)
35. Rasoul Jahanshahi, **Adam Doupé**, and Manuel Egele, "You shall not pass: Mitigating SQL Injection Attacks on Legacy Web Applications," in *Proceedings of the ACM Symposium on Information, Computer and Communications Security (AsiaCCS)*, June, 2020.
34. Christopher Salls, Aravind Machiry, **Adam Doupé**, Yan Shoshitaishvili, Christopher Kruegel, and Giovanni Vigna, "Exploring Abstraction Functions in Fuzzing," in *Proceedings of the IEEE Conference on Communications and Network Security (CNS)*, June, 2020.
33. Haehyun Cho, Jinbum Park, Donguk Kim, Ziming Zhao, Yan Shoshitaishvili, **Adam Doupé**, and Gail-Joon Ahn, "SmokeBomb: Effective Mitigation Against Cache Side-channel Attacks on the ARM Architecture," in *Proceedings of the Annual International Conference on Mobile Systems, Applications, and Services (MobiSys)*, Toronto, Canada, June 2020. (19% acceptance rate.)
32. Faris Bugra Kokulu, Ananta Soneji, Tiffany Bao, Yan Shoshitaishvili, Ziming Zhao, **Adam Doupé**, and Gail-Joon Ahn, "Matched and Mismatched SOCs: A Qualitative Study on Security Operations Center Issues," in *Proceedings of the ACM Conference on Computer and Communications Security (CCS)*, London, UK, November, 2019. (16% acceptance rate.)
31. Huahong Tu, **Adam Doupé**, Ziming Zhao, and Gail-Joon Ahn, "Users Really Do Answer Telephone Scams," in *Proceedings of the USENIX Security Symposium (USENIX)*, Santa Clara, CA, August 2019. (16% acceptance rate.)

Distinguished Paper Award

30. Erik Trickel, Oleksii Starov, Alexandros Kapravelos, Nick Nikiforakis, and **Adam Doupé**, "Everyone is Different: Client-side Diversification for Defending Against Extension Fingerprinting," in *Proceedings of the USENIX Security Symposium (USENIX)*, Santa Clara, CA, August 2019. (16% acceptance rate.)
29. Adam Oest, Yeganeh Safaei, **Adam Doupé**, Gail-Joon Ahn, Brad Wardman, and Kevin Tyers, "PhishFarm: A Scalable Framework for Measuring the Effectiveness of Evasion Techniques Against Browser Phishing Blacklists," in *Proceedings of the IEEE Symposium on Security and Privacy*, San Francisco, CA, May 2019. (12% acceptance rate.)
28. Penghui Zhang, Haehyun Cho, Ziming Zhao, **Adam Doupé**, and Gail-Joon Ahn, "iCORE: Continuous and Proactive Extrospection on Multi-core IoT Devices," in *Proceedings of the ACM Symposium on Applied Computing (SAC)*, Limassol, Cyprus, April 2019. (24% acceptance rate.)
27. Zhibo Sun, Carlos E. Rubio-Medrano, Ziming Zhao, Tiffany Bao, **Adam Doupé**, and Gail-Joon Ahn, "Understanding and Detecting Private Interactions in Underground Forums," in *Proceedings*

of the ACM on Conference on Data and Application Security and Privacy (CODASPY), Dallas, TX, March 2019.

26. Haehyun Cho, Penghui Zhang, Donguk Kim, Jinbum Park, Choong-Hoon Lee, Ziming Zhao, **Adam Doupé**, and Gail-Joon Ahn, “Prime+Count: Novel Cross-world Covert Channels on ARM TrustZone,” in *Proceedings of the Annual Computer Security Applications Conference (ACSAC)*, Puerto Rico, December 2018.
25. Jaejong Baek, Sukwha Kyung, Haehyun Cho, Ziming Zhao, Yan Shoshitaishvili, **Adam Doupé**, and Gail-Joon Ahn, “Wi Not Calling: Practical Privacy and Availability Attacks in Wi-Fi Calling,” in *Proceedings of the Annual Computer Security Applications Conference (ACSAC)*, Puerto Rico, December 2018.
24. Vaibhav Hemant Dixit, **Adam Doupé**, Yan Shoshitaishvili, Ziming Zhao, and Gail-Joon Ahn, “AIM-SDN: Attacking Information Mismanagement in SDN-datastores,” in *Proceedings of the ACM Conference on Computer and Communications Security (CCS)*, Toronto, Canada, October 2018. (16.6% acceptance rate.)
23. Mike Mabey, **Adam Doupé**, Ziming Zhao, and Gail-Joon Ahn, “Challenges, Opportunities and a Framework for Web Environment Forensics,” in *Proceedings of the IFIP International Conference on Digital Forensics*, August 2018.
22. Adam Oest, Yeganeh Safaei, **Adam Doupé**, Gail-Joon Ahn, Brad Wardman, and Gary Warner, “Inside a Phisher’s Mind: Understanding the Anti-phishing Ecosystem Through Phishing Kit Analysis,” in *Proceedings of the Symposium on Electronic Crime Research (eCrime)*, San Diego, May 2018.
21. Sai Prashanth Chandramouli, Pierre-Marie Bajan, Christopher Kruegel, Giovanni Vigna, Ziming Zhao, **Adam Doupé**, and Gail-Joon Ahn, “Measuring E-Mail Header Injections on the World Wide Web,” in *Proceedings of the ACM Symposium on Applied Computing (SAC)*, Pau, France, April 2018.
20. Sukwha Kyung, Wonkyu Han, Naveen Tiwari, Vaibhav Dixit, Lakshmi Srinivas, Ziming Zhao, **Adam Doupé**, and Gail-Joon Ahn, “HoneyProxy: Design and Implementation of Next-Generation Honeynet via SDN,” in *Proceedings of the IEEE Conference on Communications and Network Security (CNS)*, Las Vegas, NV, October 2017.
19. Sailik Sengupta, Satya Gautam Vadlamudi, Subbarao Kambhampati, **Adam Doupé**, Ziming Zhao, Marthony Taguinod, and Gail-Joon Ahn, “A Game Theoretic Approach of Strategy Generation for Moving Target Defense in Web Applications,” in *Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, Sao Paulo, May 2017.
18. Niall McLaughlin, Jesus Martinez del Rincon, BooJoong Kang, Suleiman Yerima, Paul Miller, Sakir Sezer, Yeganeh Safaei, Erik Trickel, Ziming Zhao, **Adam Doupé**, Gail-Joon Ahn, “Deep Android Malware Detection,” in *Proceedings of the ACM on Conference on Data and Application Security and Privacy (CODASPY)*, Scottsdale, Arizona, March 2017.
17. Huahong Tu, **Adam Doupé**, Ziming Zhao, and Gail-Joon Ahn, “Toward Authenticated Caller ID Transmission: The Need for a Standardized Authentication Scheme in Q.731.3 Calling Line

Identification Presentation," in *Proceedings of the ITU Kaleidoscope*, Bangkok, Thailand, November 2016.

Best Paper Award

16. Mike Mabey, **Adam Doupé**, Ziming Zhao, and Gail-Joon Ahn, "dbling: Identifying Extensions Installed on Encrypted Web Thin Clients," in *Proceedings of the Digital Forensics Research Conference (DFRWS)*, Seattle, WA, August 2016.
15. Wonkyu Han, Hongxin Hu, Ziming Zhao, **Adam Doupé**, Gail-Joon Ahn, Kuang-Ching Wang and Juan Deng, "State-aware Network Access Management for Software-Defined Networks," in *Proceedings of the ACM Symposium on Access Control Models and Technologies (SACMAT)*, Shanghai, China, June 2016.
14. Kevin Liao*, Ziming Zhao, **Adam Doupé**, and Gail-Joon Ahn, "Behind Closed Doors: Measurement and Analysis of CryptoLocker Ransoms in Bitcoin," in *Proceedings of the Symposium on Electronic Crime Research (eCrime)*, Toronto, Canada, June 2016.
13. Huahong Tu, **Adam Doupé**, Ziming Zhao, and Gail-Joon Ahn, "SoK: Everyone Hates Robocalls: A Survey of Techniques against Telephone Spam," in *Proceedings of the IEEE Symposium on Security and Privacy*, San Jose, CA, May 2016. (13.3% acceptance rate.)
12. Yiming Jing, Gail-Joon Ahn, **Adam Doupé**, and Jeong Hyun Yi, "Checking Intent-based Communication in Android with Intent Space Analysis," in *Proceedings of the ACM Symposium on Information, Computer and Communications Security (AsiaCCS)*, Xi'an, China, May 2016. (20.9% acceptance rate.)
11. Satya Gautam Vadlamudi, Sailik Sengupta, Marthony Taguinod, Ziming Zhao, **Adam Doupé**, Gail-Joon Ahn and Subbarao Kambhampati, "Moving Target Defense for Web Applications using Bayesian Stackelberg Games (Extended Abstract)," in *Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, Singapore, May 2016.
10. Vitor Afonso, Antonio Bianchi, Yanick Fratantonio, **Adam Doupé**, Mario Polino, Paulo de Geus, Christopher Kruegel, and Giovanni Vigna, "Going Native: Using a Large-Scale Analysis of Android Apps to Create a Practical Native-Code Sandboxing Policy," in *Proceedings of the ISOC Network and Distributed System Security Symposium (NDSS)*, San Diego, CA, February 2016. (15.4% acceptance rate.)
9. Carlos E. Rubio-Medrano, Ziming Zhao, **Adam Doupé**, and Gail-Joon Ahn, "Federated Access Management for Collaborative Network Environments: Framework and Case Study," in *Proceedings of the ACM Symposium on Access Control Models and Technologies (SACMAT)*, Vienna, Austria, June 2015. (28.8% acceptance rate.)
8. Yan Shoshitaishvili, Luca Invernizzi, **Adam Doupé**, and Giovanni Vigna, "Do You Feel Lucky? A Large-Scale Analysis of Risk-Rewards Trade-Offs in Cyber Security," in *Proceedings of the ACM Symposium on Applied Computing (SAC 2014)*, Gyeongju, Korea, March 2014.
7. **Adam Doupé**, Weidong Cui, Mariusz H. Jakubowski, Marcus Peinado, Christopher Kruegel, and Giovanni Vigna, "deDacota: Toward Preventing Server-Side XSS via Automatic Code and Data

Separation,” in *Proceedings of the 20th ACM Conference on Computer and Communications Security (CCS)*, Berlin, Germany, November 2013. (19.8% acceptance rate.)

6. **Adam Doupé** and Janet L. Kayfetz, “Writing Groups in Computer Science Research Labs,” in *Proceedings of the Frontiers in Education Conference (FIE 2013)*, Oklahoma City, OK, October 2013.
5. Pierre Payet, **Adam Doupé**, Christopher Kruegel, and Giovanni Vigna, “EARs in the Wild: Large-Scale Analysis of Execution After Redirect Vulnerabilities,” in *Proceedings of the ACM Symposium on Applied Computing (SAC 2013)*, Coimbra, Portugal, March 2013.
4. **Adam Doupé**, Ludovico Cavedon, Christopher Kruegel, and Giovanni Vigna, “Enemy of the State: A State-Aware Black-Box Web Vulnerability Scanner,” in *Proceedings of the 21st USENIX conference on Security (USENIX 2012)*, Bellevue, WA, August 2012. (19.4% acceptance rate.)

Outstanding Publication Award in Computer Science, UC Santa Barbara

3. **Adam Doupé**, Manuel Egele, Benjamin Caillat, Gianluca Stringhini, Gorkem Yakin, Ali Zand, Ludovico Cavedon, and Giovanni Vigna, “Hit ‘em Where it Hurts: A Live Security Exercise on Cyber Situational Awareness,” in *Proceedings of the Annual Computer Security Applications Conference (ACSAC 2011)*, Austin, TX, December 2011. (20% acceptance rate.)
2. **Adam Doupé**, Bryce Boe, Christopher Kruegel, and Giovanni Vigna, “Fear the EAR: Discovering and Mitigating Execution After Redirect Vulnerabilities,” in *Proceedings of the 18th ACM Conference on Computer and Communications Security (CCS 2011)*, Chicago, IL, October 2011. (14% acceptance rate.)
1. **Adam Doupé**, Marco Cova, and Giovanni Vigna, “Why Johnny Can’t Pentest: An Analysis of Black-box Web Vulnerability Scanners,” in *Proceedings of the Seventh Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA 2010)*, Bonn, Germany, July 2010. (34% acceptance rate.)

Workshop Papers with Proceedings (refereed)

11. Sana Habib, Tiffany Bao, Yan Shoshitaishvili, and **Adam Doupé**, “Mitigating Threats Emerging from the Interaction between SDN Apps and SDN (Configuration) Datastore,” in *Proceedings of the ACM Cloud Computing Security Workshop (CCSW)*, November, 2022.
10. Haehyun Cho, Jinbum Park, Joonwon Kang, Tiffany Bao, Ruoyu (Fish) Wang, Yan Shoshitaishvili, **Adam Doupé**, and Gail-Joon Ahn, “Exploiting Uses of Uninitialized Stack Variables in Linux Kernels to Leak Kernel Pointers,” in *Proceedings of the USENIX Workshop on Offensive Technologies (WOOT)*, August, 2020.
9. Mauricio Gutierrez, Ziming Zhao, **Adam Doupé**, Yan Shoshitaishvili, and Gail-Joon Ahn, “Cache-Light: Defeating the CacheKit Attack,” in *Proceedings of the Workshop on Attacks and Solutions in Hardware Security (ASHES)*, Toronto, Canada, October 2018.
8. Vaibhav Dixit, Sukwha Kyung, Ziming Zhao, **Adam Doupé**, Yan Shoshitaishvili, and Gail-Joon Ahn, “Challenges and Preparedness of SDN-based Firewalls,” in *Proceedings of the ACM International*

Workshop on Security in Software Defined Networks and Network Function Virtualization (SDN-NFV Sec), Tempe, AZ, March 2018.

7. Carlos E. Rubio-Medrano, Josephine Lamp*, **Adam Doupé**, Ziming Zhao, and Gail-Joon Ahn, "Mutated Policies: Towards Proactive Attribute-based Defenses for Access Control," in *Proceedings of the Workshop on Moving Target Defense (MTD)*, Dallas, TX, October 2017.
6. Erik Trickel, Francesco Disperati, Eric Gustafson, Faezeh Kalantari, Mike Mabey, Naveen Tiwari, Yeganeh Safaei, **Adam Doupé**, and Giovanni Vigna, "Shell We Play A Game? CTF-as-a-service for Security Education," in *Proceedings of the USENIX Workshop on Advances in Security Education (ASE)*, Vancouver, CA, August 2017.
5. Patrick Mutchler, Yeganeh Safaei, **Adam Doupé**, and John Mitchell, "Target Fragmentation in Android Apps," in *Proceedings of the Mobile Security Technologies Workshop (MoST)*, San Jose, CA, May 2016.
4. Wonkyu Han, Ziming Zhao, **Adam Doupé**, and Gail-Joon Ahn, "HoneyMix: Toward SDN-based Intelligent HoneyNet," in *Proceedings of the ACM International Workshop on Security in Software Defined Networks and Network Function Virtualization (SDN-NFV Sec)*, New Orleans, LA, March 2016.
3. Carlos E. Rubio-Medrano, Josephine Lamp*, Marthony Taguinod, **Adam Doupé**, Ziming Zhao, and Gail-Joon Ahn, "Position Paper: Towards a Moving Target Defense Approach for Attribute-based Access Control," in *Proceedings of the ACM Workshop on Attribute Based Access Control (ABAC)*, New Orleans, LA, March 2016.
2. Patrick Mutchler, **Adam Doupé**, John Mitchell, Christopher Kruegel, and Giovanni Vigna, "A Large-Scale Study of Mobile Web App Security," in *Proceedings of the Mobile Security Technologies Workshop (MoST)*, San Jose, CA, May 2015.

Best Paper Award

1. Giovanni Vigna, Kevin Borgolte, Jacopo Corbetta, **Adam Doupé**, Yanick Fratantonio, Luca Invernizzi, Dhilung Kirat, and Yan Shoshitaishvili, "Ten Years of iCTF: The Good, The Bad, and The Ugly," in *Proceedings of the USENIX Summit on Gaming, Games and Gamification in Security Education (3GSE)*, San Diego, CA, August 2014.

Book Chapters

1. Gail-Joon Ahn, **Adam Doupé**, Ziming Zhao, and Kevin Liao*, "Ransomware and cryptocurrency: Partners in crime," book chapter in Thomas J. Holt, Ed., *Cybercrime Through an Interdisciplinary Lens*, pp. 105–126, Routledge Studies in Crime and Society, Taylor & Francis Publishers, 2016.

Invited Papers

2. Ajay Modi, Zhibo Sun, Anupam Panwar, Tejas Khairnar, Ziming Zhao, **Adam Doupé**, and Gail-Joon Ahn, "Towards Automated Threat Intelligence Fusion," in *Proceedings of the IEEE International Conference on Collaboration and Internet Computing (CIC)*, November 2016.

1. Marthony Taguinod, **Adam Doupé**, Ziming Zhao, and Gail-Joon Ahn, "Toward a Moving Target Defense for Web Applications," in *Proceedings of the IEEE International Conference on Information Reuse and Integration (IRI)*, San Francisco, CA, August 2015.

Patent Activity

2. Huahong Tu, **Adam Doupé**, Ziming Zhao, and Gail-Joon Ahn, "Systems and Methods of Authenticating Caller Identity and Call Request Header Information for Outbound Telephony Communications," *Patent US10447481B2*, 2019.
1. Weidong Cui, **Adam Doupé**, Mariusz Jakubowski, and Marcus Peinado, "Automatic Code and Data Separation of Web Application," *Patent US9774620B2*, 2017.

Invited Talks

22. "Wild Wild Phish: Phrontiers in the Phight Against Phishing" Keynote at the SecWeb Workshop, May, 2022.
21. "Scam Pandemic: How Attackers Exploit Public Fear through Phishing," Invited talk for the FTC, July, 2021.
20. "Cybersecurity, ASU, and You," Invited talk for the GenCyber camp for High Schoolers, June, 2021.
19. "Where Have All the Phishers Gone?" Keynote at the Workshop on Measurements, Attacks, and Defenses for the Web (MADWeb), February, 2021.
18. Moderated the "Scaling the Academic Security Community" panel at the ACM Conference on Computer and Communications Security, November, 2019.
17. "History and Future of Automated Vulnerability Analysis," Junior Keynote at ACM Symposium on Access Control Models and Technologies (SACMAT), June 2019.
16. Participated in the "Browsers and Security" panel at the Workshop on Measurements, Attacks and Defenses for the Web (MADWeb), February 2019.
15. "Event-driven Models and Security Research," Invited joint talk (with Fish Wang) for NETCOM Command Strategy Group, Sierra Vista, AZ, January 2019.
14. "The Effectiveness of Telephone Phishing Scams and Possible Solutions," Invited talk at University of Illinois at Chicago, Chicago, IL, January 2018.
13. "Hack like a Human: Using Inductive Reverse Engineering for Black-Box Web Vulnerability Analysis," Invited talk at UC Santa Barbara, Santa Barbara, CA, November 2017.
12. Participated in the "Education and Workforce" panel at the ASU Congressional Conference on Cybersecurity, August 2017.
11. Participated in the "Rigor in Experimentation" panel at the USENIX Workshop on Cyber Security Experimentation and Test (CSET), August 2017.

10. "The Effectiveness of Telephone Phishing Scams and Possible Solutions," Invited talk at University College London, London, UK, June 2017.
9. "Everybody Hates Robocalls: What Can We Do About It?," Invited talk at Stony Brook University, Stony Brook, NY, April 2017.
8. Moderated the "Trustworthy Data Science" panel at the ACM Conference on Data and Applications Security and Privacy (CODASPY), March 2017.
7. "Large-Scale Security Analysis of Android Applications," Invited talk for the 3rd International Symposium on Mobile Security (MSEC 2016), December 2016.
6. Moderated the "Security Competitions" panel at the USENIX Security Symposium, August 2016.
5. "A Computer in Every Pocket: Securing Mobile Applications," Invited talk by Dr. Jennifer Hackney Price, ASU West, September 2016.
4. "Everything You've Ever Wanted to Know About Black-Box Web Vulnerability Scanners (But Were Afraid to Ask)," Invited talk, Open Web Application Security Project (OWASP) Phoenix Chapter, June 2016.
3. "Using Static Analysis to Tame the Wild Wild Web," Invited talk at Georgia Tech, Georgia, AL, February 2015.
2. "Into the Mind of the Hacker," Three hour workshop at UC Santa Barbara by request of Web Standard Group, Santa Barbara, CA, April 2012.
1. "Fear the EAR: Discovering and Mitigating Execution After Redirect Vulnerabilities," Invited talk at Raytheon, Santa Barbara, CA, October 2011.

Service

Program Committee

- 2025 IEEE Symposium on Security and Privacy (Oakland)
- 2025 Workshop on Binary Analysis Research (BAR)
- 2024 ACM Conference on Computer and Communications Security (CCS)
- 2024 International Symposium on Research in Attacks, Intrusions and Defenses (RAID)
- 2024 USENIX Security Symposium (USENIX)
- 2024 IEEE Symposium on Security and Privacy (Oakland)
- 2024 Workshop on Binary Analysis Research (BAR)
- 2022 ACM Conference on Computer and Communications Security (CCS)
- 2022 USENIX Security Symposium (USENIX)
- 2022 Workshop on Measurements, Attacks, and Defenses for the Web (MADWeb)
- 2022 Network and Distributed System Security Symposium (NDSS)

2021 The Web Conference (WWW)
 2021 European Workshop on Systems Security (EuroSec)
 2021 Workshop on Measurements, Attacks, and Defenses for the Web (MADWeb)
 2021 Network and Distributed System Security Symposium (NDSS)
 2021 USENIX Security Symposium (USENIX)
 2021 IEEE Symposium on Security and Privacy
 2020 Annual Computer Security Applications Conference (ACSAC)
 2020 SecWeb Workshop (SecWeb)
 2020 ACM Conference on Computer and Communications Security (CCS)
 2020 Workshop on Measurements, Attacks, and Defenses for the Web (MADWeb)
 2020 Network and Distributed System Security Symposium (NDSS)
 2020 USENIX Security Symposium (USENIX)
 2020 The Web Conference (WWW)
 2020 ACM Symposium on Information, Computer and Communications Security (AsiaCCS)
 2019 APWG Annual Symposium on Electronic Crime Research (eCrime)
 2019 Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA)
 2019 ACM Conference on Computer and Communications Security (CCS)
 2020 IEEE Symposium on Security and Privacy
 2019 ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)
 2019 USENIX Security Symposium (USENIX)
 2019 IEEE Workshop on the Internet of Safe Things (SafeThings)
 2019 The Web Conference (WWW)
 2019 Workshop on Measurements, Attacks, and Defenses for the Web (MADWeb)
 2019 Network and Distributed System Security Symposium (NDSS)
 2018 Annual Computer Security Applications Conference (ACSAC)
 2018 ACM Conference on Computer and Communications Security (CCS)
 2019 IEEE Symposium on Security and Privacy
 2018 Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA)
 2018 APWG Annual Symposium on Electronic Crime Research (eCrime)
 2018 IEEE International Conference on Distributed Computing Systems (ICDCS)
 2018 USENIX Security Symposium (USENIX)
 2018 Network and Distributed System Security Symposium (NDSS)
 2017 Annual Computer Security Applications Conference (ACSAC)
 2017 ACM Conference on Computer and Communications Security (CCS)

2017 Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA)
2017 USENIX Security Symposium (USENIX)
2017 ACM Symposium on Applied Computing (SAC)
2016 ACM Conference on Computer and Communications Security (CCS)
2016 USENIX Security Symposium (USENIX)
2016 International Conference on Availability, Reliability and Security (ARES)
2016 International World Wide Web Conference (WWW)
2016 ACM Symposium on Applied Computing (SAC)
2015 ACM Conference on Computer and Communications Security (CCS)
2015 USENIX Security Symposium (USENIX)
2015 International Conference on Availability, Reliability and Security (ARES)
2015 International Conference on Web Engineering (ICWE)
2015 OWASP AppSecEU
2015 International World Wide Web Conference (WWW)
2013 GreHack

Conference Reviewer

2016 Frontiers in Education (FIE)
2015 Frontiers in Education (FIE)
2014 Frontiers in Education (FIE)
2013 Frontiers in Education (FIE)

Journal Reviewer

- ACM Transactions on Architecture and Code Optimization
- ACM Transactions on the Web
- ACM Transactions on Information and System Security (TISSEC)
- Computers — Open Access Journal
- Future Generation Computer Systems
- IET Software
- IEEE Transactions on Information Forensics & Security
- IEEE Transactions on Dependable and Secure Computing
- IEEE Transactions on Reliability
- IEEE/ACM Transactions on Networking
- Journal of Software Testing, Verification and Reliability (STVR)

Journal Organization

2020–2024 Associate Editor of IEEE Transactions on Dependable and Secure Computing (TDSC)

Conference Organization

- 2025 Program Chair for the Hot Topics in the Science of Security (HotSoS) Symposium
- 2025 Area Chair of The Web Conference (WWW)
- 2024 Program Co-Chair of the USENIX WOOT Conference on Offensive Technologies (WOOT)
- 2024 Area Chair of The Web Conference (WWW)
- 2023 Associate Chair of the IEEE Symposium on Security and Privacy
- 2021 Web Security Track Chair for the ACM Conference on Computer and Communications Security (CCS)
- 2019 Program co-Chair for the Workshop on Forming an Ecosystem Around Software Transformation (FEAST)
- 2019 Panel Chair for the ACM Conference on Computer and Communications Security (CCS)
- 2019 Student Travel Grant Committee for the Network and Distributed System Security Symposium (NDSS)
- 2018 Student Travel Grant Committee for the Network and Distributed System Security Symposium (NDSS)
- 2018 Tutorial Chair for ACM Symposium on Information, Computer and Communications Security (AsiaCCS)
- 2018 Distinguished Paper Committee for USENIX Security Symposium (USENIX)
- 2017 Panel Chair for ACM Conference on Data and Application Security and Privacy (CODASPY)
- 2015 Poster Co-Chair for USENIX Security Symposium
- 2012 Chair of Graduate Student Workshop on Computing (GSWC), UC Santa Barbara. Raised record \$8,600 from industry, organized larger venue, ensured reviews were written, facilitated PC meeting, and hosted actual event.
- 2011 Industry Liaison for Graduate Student Workshop on Computing (GSWC), UC Santa Barbara. Raised record \$8,500 from industry sponsors Google, AppFolio, Lastline, Qualcomm, and Microsoft.

Proposal Reviewer

- 2024 NSF SaTC Panel Reviewer
- 2021 NSF SaTC Panel Reviewer
- 2020 NSF SaTC Panel Reviewer
- 2019 NSF SaTC Panel Reviewer
- 2018 NSF SaTC Panel Reviewer
- 2018 NSF SaTC-EDU Panel Reviewer

2015 NSF SaTC Panel Reviewer

Faculty Advisor for Student Groups

2014–Present Faculty Advisor for ASU Hacking Club (formerly known as pwndevils), a student organization affiliated with the Center for Cybersecurity and Trusted Foundations. The group meets weekly to practice ethical hacking skills to compete in Capture the Flag competitions.

2016–2022 Faculty Advisor for AZ Saber, a student group dedicated to teaching people the basics of swordsmanship and safety.

2015–2022 Faculty Advisor for Software Developers Association (SoDA), the largest student organization in the Fulton Schools of Engineering

Department Service

2024–Present Personnel Committee, Arizona State University

2024 Trustworthy and Secure Social Connectivity Search Committee **Chair**, Arizona State University

2023–Present Graduate Program Committee, Arizona State University

2022 Cybersecurity, Computation, and Consensus Search Committee **Chair**, Arizona State University

2019–2022 Undergraduate Program Committee, Arizona State University

2018 Cybersecurity Faculty Search Committee **Chair**, Arizona State University

2014–2018 PhD Admissions Committee, Arizona State University

2016 Cybersecurity Faculty Search Committee, Arizona State University

2015 Cybersecurity Faculty Search Committee, Arizona State University

Advisory Board

2017–2019 Advisory board of the EPSRC UK project “ACCEPT: Addressing Cybersecurity and Cybercrime via a co-Evolutionary aPproach to reducing human-relaTed risks” <http://accept.sccs.surrey.ac.uk>

Media Interactions

January, 14, “ASU cybercrime-fighter helps us compute with confidence.” Interviewed in ASU News
2025 Q&A about the Center for Cybersecurity and Trusted Foundations’ mission and recent successes.

September, 9, “Cybersecurity team including students, faculty from ASU wins \$2 million in inter-
2024 national contest.” Live interview with KTAR News (92.3 FM) about the Shellphish qualifying for DARPA’s AIXCC competition.

- October, 24, 2023 "Cybersecurity threats that keep experts up at night." Featured in ASU News Q&A interview for Cybersecurity Awareness Month discussing online vigilance and emerging threats like e-commerce scam sites.
- January, 31, 2022 "Fortifying the foundations of cybersecurity." Quoted in ASU News article announcing the launch of ASU's new cybersecurity center, explaining the goal of keeping users and their data safe through research and education.
- July, 28, 2021 "Equipping the next generation of cybersecurity professionals." Quoted in ASU News article about preparing the next generation of cybersecurity professionals at ASU.
- August, 27, 2020 "Zoom bombings are still happening during virtual school. Why?" Interviewed by Lily Altavena of Arizona Republic.
- August, 6, 2019 "Hacking with elite white hats." Interviewed by Scott Seckel on DEF CON CTF.
- February, 6, 2019 "Especialistas alertam para ligações robôs." Interviewed by GloboTV for Brazilian news piece on robocalls.
- September, 13, 2018 "ASU cybersecurity professionals encourage students to join the field." Interviewed by the State Press regarding cybersecurity jobs.
- April, 4, 2018 "You Can't Clean Up a Data Spill." Quoted by April Glaser for Slate on the Facebook data leaks.
- October, 10, 2017 "ASU director testifies at Equifax Senate hearing." Interviewed by Ethan Millman for The State Press on the Equifax data breach.
- August, 28, 2017 "Junk call nightmare flooded woman with hundreds of bizarre phone calls a day." Interviewed by Jon Brodtkin for Ars Technica on our robocall and caller ID spoofing research.
- August, 15, 2017 "How you can stop those annoying robocalls." Appearance on CBS WNCN (North Carolina) discussing our robocall research.
- August, 7, 2017 "ASU engineer Adam Doupé looks to safeguard personal data through advanced automated vulnerability analysis tools." Interview with Erik Wirtanen on my CAREER award
- June, 19, 2017 "ASU team tests 'green padlock' solution to scam phone calls." Interviewed by Derek Staahl for 3TV and CBS 5 for a story on our caller ID security indicator research
- June, 13, 2017 "Receiving more robocalls than ever? An ASU researcher may have a solution." Interview with Leslie Minton on our telephony security research
- March, 8, 2017 "Robocall ringing your cellphone? Here's what not to do." My PhD student Huahong (Raymond) Tu interviewed by Ian Schwartz for CBS 5 for a story on robocalls
- January, 31, 2017 "Professors explain the flaws of a 'password system' in a digital world." Interview by Aravind Sreenivasa for The State Press, ASU's student paper, on the state of password security
- January, 10, 2017 "State computers hit with suspected malware attack." Interview with William Pitts for 12 News about recent reports of state of AZ HR computers infected with malware

- October, 19, 2016 "How the FCC, the telecom industry and computer programmers are working to fight robocalls." My PhD student Huahong (Raymond) Tu interviewed by Larry Mantle for LA's KPCC's AirTalk radio program for a story on robocalls
- October, 4, 2016 "Fed up with rising robocalls, millions say 'Do Not Call' list doesn't work and want relief." My PhD student Huahong (Raymond) Tu quoted in The Orange Country Register for a story on robocalls by Lily Leung
- September, 23, 2016 "Why is Yahoo hack so problematic?." Interview with Ryan Cody for 12 News about the impact of the Yahoo breach and why email security is so important.
- August, 29, 2016 "Russian hack targets Arizona voter information." Interview with Brahm Resnik for 12 News about recent allegations of Russian hackers targeting Arizona voter registration.
- August, 29, 2016 "Russian hackers tried to break into AZ voter database." Interview with Brahm Resnik for 12 News about recent allegations of Russian hackers targeting Arizona voter registration.
- July, 27, 2016 "DNC email leak should be a lesson for all of us." Interview with Ryan Cody for 12 News about the importance of securing your personal email and tips to keep your personal email secure.
- March, 1, 2016 "Software Development Association students compete, network and have fun." Story about the ASU Software Developer Association (SoDA) 2016 Coding Competition.
- February, 4, 2016 "Arizona Department of Revenue IT Systems Vulnerable." Interview with Amber Kahwaji for Cronkite News about an AZ Department of Revenue Report on IT Vulnerabilities
- July, 27, 2015 "Exclusive: MCAO talked about buying spyware." Interview with William Pitts for 12 News about details of the Hacking Team leaks.
- April, 22, 2015 "Feds: Terrorist could hack your plane." Interview with William Pitts for 12 News about hacking airplanes.

Research Experience

- 2013 **Visiting PhD Student**, *Stanford University*, Advisor: John C. Mitchell
(Summer)
- 2012 **Research Intern**, *Microsoft Research*, Advisor: Weidong Cui
(Summer)

Industry Experience

- 2009–2010 **Software Developer Engineer**, *Microsoft*, Developed as a team a C# Microsoft Word plug-in to assist technical writers. Met critical deadlines in high-pressure situations
- 2008 **Software Developer Engineer Intern**, *Microsoft*, Developed a C# application that
(Summer) could save a technical writer documenting XML schemas 20 hours in 2 weeks
- 2007–2008 **Software Developer**, *AT&T Government Solutions*, Developed and maintained a large C++ desktop application

- 2005–2010 **Founder/Developer**, *WootWatchers.com*, Founded WootWatchers.com using Ruby on Rails connecting to a MySQL backend. At its peak had 4,000 users
- 2004–2005 **Mobile Product Manager**, *VCEL, Inc.*, Started off developing a client-server J2ME application. Then maintained and developed the server back-end using PHP

--- Capture the Flag Organization

- 2018–2022 **Founding Member**, *Order of the Overflow*, Lead a group of hackers to organize the DEF CON Capture the Flag (CTF) competition, considered the “Olympics” of CTF. We host two competitions annually: DEF CON CTF Quals (which teams from all around the world play), and where the top teams from that competition compete in DEF CON CTF Finals, co-located with the DEF CON security conference.

--- Entrepreneurship

- 2024–Present **Co-founder**, *Pwn In Motion, LLC*, Productize open-source research prototypes for clients.
- 2023–Present **Co-founder**, *pwn.college, LLC*, Provide educational cybersecurity trainings. Finished Phase I STTR and currently in Phase II STTR, 30% of which goes to ASU.

--- Teaching Experience

Instructor

- Fall 2024 CSE 466 “Computer Systems Security” at ASU. **4.51**/5 course evaluation and **4.7**/5 instructor evaluation from 7 of 50 enrolled students.
- Fall 2024 CSE 466 “Computer Systems Security” at ASU. **5**/5 course evaluation and **5**/5 instructor evaluation from 2 of 16 enrolled students.
- Fall 2024 CSE 598 “Applied Vulnerability Research” at ASU. **4.69**/5 course evaluation and **5**/5 instructor evaluation from 3 of 13 enrolled students.
- Spring 2024 CSE 598 “Advanced Software Exploitation” at ASU. **4.94**/5 course evaluation and **4.92**/5 instructor evaluation from 7 of 10 enrolled students.
- Fall 2023 CSE 365 “Introduction to Information Assurance” at ASU. **4.57**/5 course evaluation and **4.58**/5 instructor evaluation from 75 of 323 enrolled students.
- Fall 2023 CSE 365 “Introduction to Information Assurance” at ASU. **4.55**/5 course evaluation and **4.58**/5 instructor evaluation from 79 of 337 enrolled students.
- Spring 2022 CSE 365 “Introduction to Information Assurance” at ASU. **4.63**/5 course evaluation and **4.72**/5 instructor evaluation from 56 of 238 enrolled students.
- Spring 2021 CSE 365 “Introduction to Information Assurance” at ASU. **4.61**/5 course evaluation and **4.78**/5 instructor evaluation from 62 of 272 enrolled students.
- Spring 2020 CSE 365 “Introduction to Information Assurance” at ASU. **4.61**/5 course evaluation and **4.77**/5 instructor evaluation from 100 of 349 enrolled students.

- Fall 2019 CSE 365 "Introduction to Information Assurance" at ASU. **4.63**/5 course evaluation and **4.72**/5 instructor evaluation from 86 of 186 enrolled students.
- Fall 2018 CSE 365 "Introduction to Information Assurance" at ASU. **4.72**/5 course evaluation and **4.87**/5 instructor evaluation from 24 of 100 enrolled students.
- Spring 2018 CSE 545 "Software Security" at ASU. **4.66**/5 course evaluation and **4.78**/5 instructor evaluation from 56 of 107 enrolled students.
- Fall 2017 CSE 465 "Information Assurance" at ASU. **4.58**/5 course evaluation and **4.74**/5 instructor evaluation from 27 of 120 enrolled students.
- Spring 2017 CSE 545 "Software Security" at ASU. **4.72**/5 course evaluation and **4.82**/5 instructor evaluation from 92 of 139 enrolled students.
- Fall 2016 CSE 340 "Principles of Programming Languages" at ASU. **4.76**/5 course evaluation and **4.8**/5 instructor evaluation from 71 of 223 enrolled students.
- Spring 2016 CSE 340 "Principles of Programming Languages" at ASU. **4.5**/5 course evaluation and **4.88**/5 instructor evaluation from 32 of 119 enrolled students.
- Spring 2016 CSE 545 "Software Security" at ASU. **4.73**/5 course evaluation and **4.79**/5 instructor evaluation from 73 of 119 enrolled students.
- Fall 2015 CSE 340 "Principles of Programming Languages" at ASU. **4.46**/5 course evaluation and **4.74**/5 instructor evaluation from 92 out of the 204 students.
- Spring 2015 Created CSE 591 "Security and Vulnerability Analysis" at ASU. **4.73**/5 course evaluation and **4.89**/5 instructor evaluation from 48 of 63 enrolled students.
- Fall 2013 Co-created and co-taught "Recent Trends in Computing Research" at University of California, Santa Barbara, a 2-unit seminar graduate class.

Current Thesis Advisor (Academic Committee Chair)

- 2015–Present Yeganeh Safaei (PhD Student)
- 2016–Present Faris Kokulu (PhD Student)
- 2019–Present Ananta Soneji (PhD Student)
- 2019–Present Wil Gibbs (PhD Student)
- 2022–Present Zhuoer Lyu (PhD Student)
- 2022–Present Zach Smith (PhD Student)
- 2024–Present Alemi Farhad (PhD Student)
- 2024–Present Giorgi Arsenadze (PhD Student)
- 2024–Present Davit Benashvili (PhD Student)

Prior PhD Advisor (Academic Committee Chair)

- 2025 Marzieh Bitaab (PhD) Dissertation: "A Study of Fraudulent E-Commerce Website Modus Operandi." Amazon.

- 2024 Faezeh Kalantari (PhD) Dissertation: "Safeguarding Users' Web Interactions." Fingerprint.
- 2024 Connor Nelson (PhD) Dissertation: "Hacking the Learning Curve: Effective Cybersecurity Education at Scale." Research Software Engineer, ASU.
- 2023 Vivin Paliath (PhD) Dissertation: "Modeling State to Improve Defensive Cyberattack Strategies." co-founder of startup Basata.
- 2023 Erik Trickel (PhD) Dissertation: "Attacking Computer Security from the Perspective of Educators, Users, and Analysts." Assistant Teaching Professor, ASU.
- 2023 Zhibo (Eric) Sun (PhD) Dissertation: "In the Light and in the Shadows: Human-Centered Analysis in Cybercrime." Assistant Professor, Drexel University.
- 2022 Penghui Zhang (PhD) Dissertation: "Treatments Against Sophisticated Phishing Attacks." Meta.
- 2021 Haehyun Cho (PhD) Dissertation: "Exploiting and Mitigating Advanced Security Vulnerabilities." Assistant Professor, Soongsil University.
- 2020 Adam Oest (PhD) Dissertation: "Leveraging Scalable Data Analysis to Proactively Bolster the Anti-Phishing Ecosystem." Paypal.
- 2017 Huahong Tu (PhD) Dissertation: "From Understanding Telephone Scams to Implementing Authenticated Caller ID Transmission." Assistant Clinical Professor, University of Maryland.
- 2017 Michael Mabey (PhD) Dissertation: "Forensic Methods and Tools for Web Environments."

Prior Masters Advisor (Academic Committee Chair)

- 2022 Purv Rakeshkumar Chauhan (Masters) Thesis: "Trapped in Transparency: Analyzing the Effectiveness of Security Defenses in Real-World Scenarios." Nextiva.
- 2020 Jaswant Pakki (Masters) Thesis: "Everything You Ever Wanted to Know About Bitcoin Mixers (But Were Afraid to Ask)." Raytheon.
- 2020 Will Gibbs (Masters) Thesis: "Natural Intent: The Use and Misuse of Intents in Android Applications."
- 2019 Zackary Crosley (Masters) Thesis: "Automated Reflection of CTF Hostile Exploits (ARCHES): Inductive Programming Techniques for Network Traffic Comprehension and Reflection."
- 2019 Connor Nelson (Masters) Thesis: "IRE: A Framework For Inductive Reverse Engineering."
- 2018 Vaibhav Hemant Dixit (Masters) Thesis: "Threats and Defenses in SDN Control Plane."
- 2018 James Keith Hutchins (Masters) Thesis: "FrozenNode: Static Linking of Node.js Applications."

- 2017 Marthony T. Taguinod (Masters) Thesis: "Moving Target Defense Framework for Web Applications."
- 2017 Ashwath Anand (Masters) Thesis: "Threats, Countermeasures, and Research Trends for BLE-Based IoT Devices."
- 2017 Gerard Pinto (Masters) Thesis: "Shadow Phone and Ghost SIM: A Step toward Geo-Location Anonymous Calling in GSM." First job at IBM.
- 2017 Tejas Khairnar (Masters) Thesis: "Next Generation Black-Box Web Application Vulnerability Analysis Framework."
- 2016 Sai Chandramouli (Masters) Thesis: "E-Mail Header Injections: An Analysis of the World Wide Web." First job at Amazon.

Prior Undergraduate Thesis Advisor

- 2017 Kevin Liao (Undergraduate) Thesis: "Toward Inductive Reverse Engineering of Web Applications." PhD Student at UIUC.